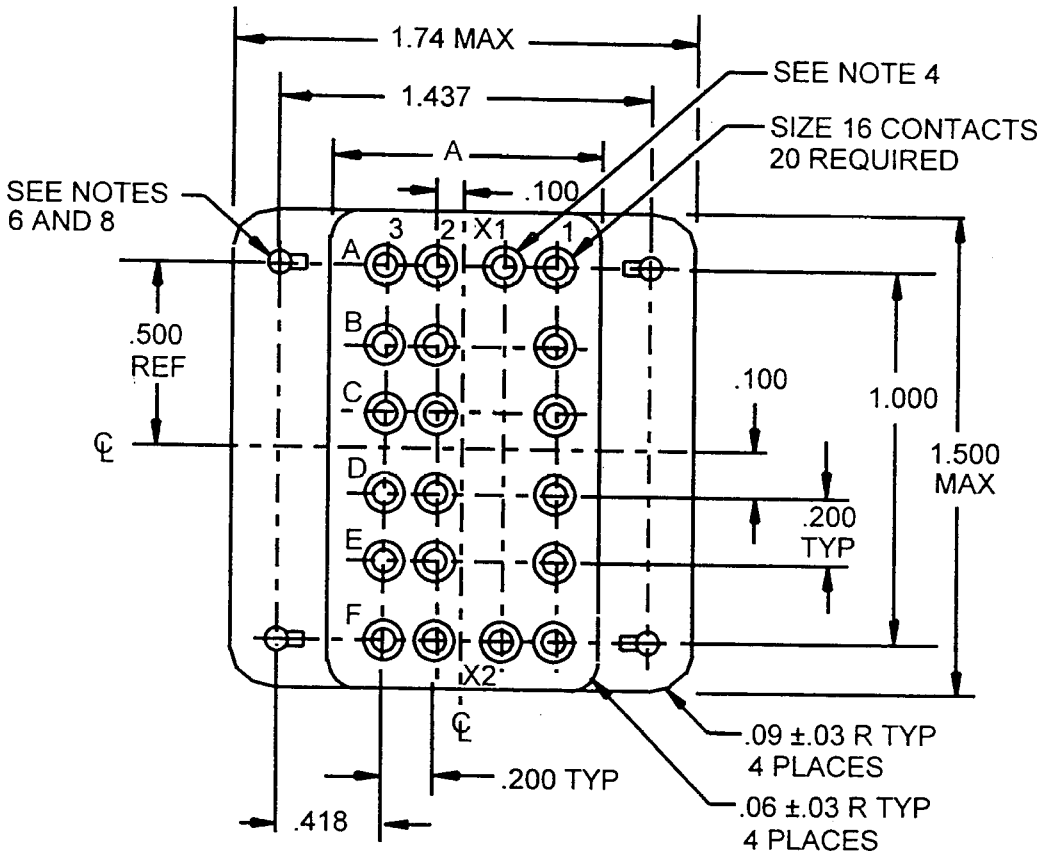
INCH-POUND
MIL-PRF-12883/47B
28 June 1996
SUPERSEDING
MIL-S-12883/47A
20 December 1989

PERFORMANCE SPECIFICATION SHEET

SOCKET, PLUG-IN ELECTRONIC COMPONENTS, FOR RELAYS, 6-POLE, 10 AMPERES (MIL-R-6106 AND MIL-R-83536)

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-12883.



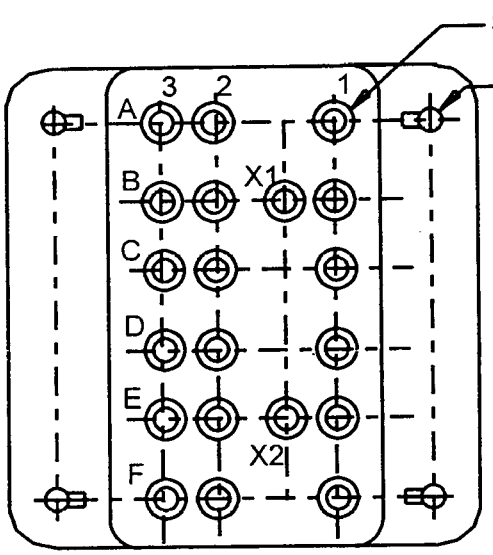
MATING FACE

-01, -04, -07, and -10

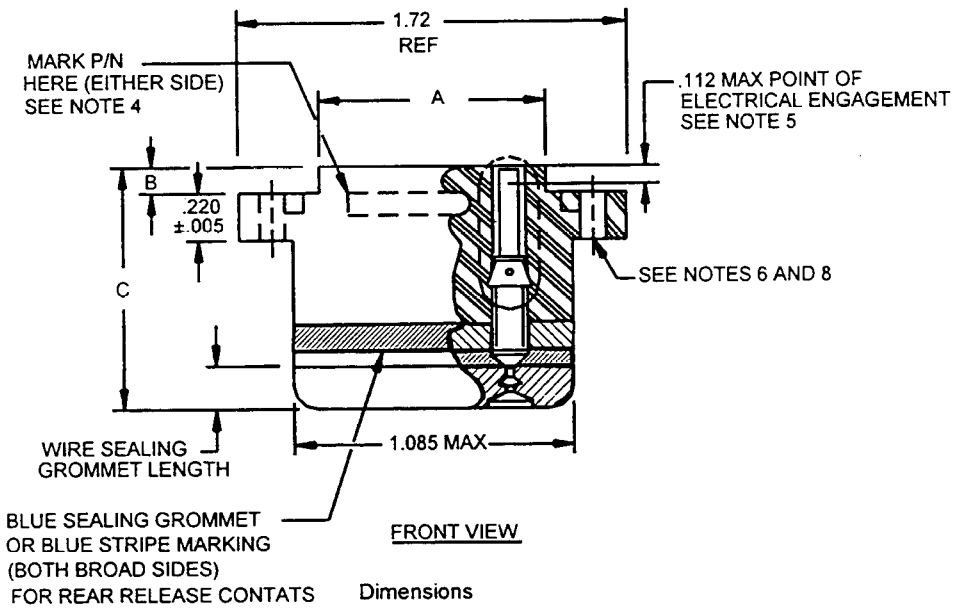
FIGURE 1. Socket.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MATING FACE
-02, -05, -08, and -11



FRONT VIEW

Dimensions

Dash number	A (mm)	B (mm)	C Max (mm)
-01, -02, -04, and -05	1.075 (27.31)	.093 (2.36)	.890 (22.61)
-07, -08, -10, and -11	1.025 (26.04)	.140 - .135 (356 - 3.43)	1.250 (31.75)

FIGURE 1. Socket configurations - Continued.

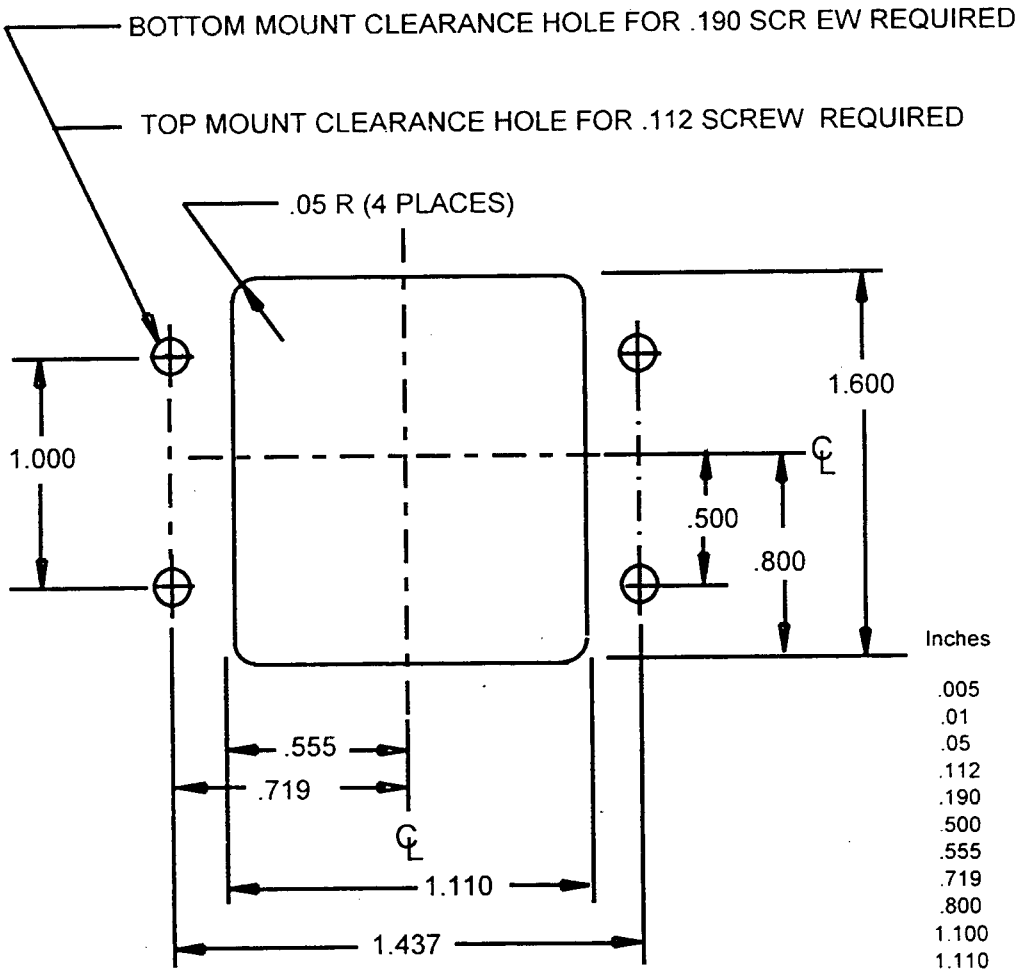
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Inches	mm	Inches	mm
.005	0.13	.418	10.62
.01	0.25	.500	12.70
.03	0.8	.512	13.00
.035	0.90	.890	22.61
.06	1.5	1.000	25.40
.09	2.3	1.025	26.04
.093	2.36	1.075	27.30
.100	2.54	1.085	27.56
.112	2.84	1.250	31.75
.135	3.43	1.437	36.50
.140	3.56	1.500	38.10
.150	3.81	1.72	43.7
.200	5.08	1.720	43.69
.220	5.59		

NOTES
1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. Unless otherwise specified tolerances are ±.01 (0.25 mm) for two place decimals and ±.005 (0.13 mm) for three place decimals.
4. Marking shall be characters which are molded .035 (0.90 mm) minimum ink marking optional per MIL-STD-1285.
5. Point of electrical contact from mating face of socket insulator to the socket contact.
6. Keyway is shown for loose stud mounting configuration only (see figure 4, detail A).
7. For mating relay (see table I).
8. Configuration for mounting (see figures 3, 4 and table I).

FIGURE 1. Socket configuration - Continued.

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Inches	mm
.005	0.13
.01	0.25
.05	1.3
.112	2.84
.190	4.83
.500	12.70
.555	14.10
.719	18.26
.800	20.32
1.100	25.40
1.110	28.19
1.437	36.50
1.600	40.64

RECOMMENDED MOUNTING DIMENSIONS

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerances are ±.005 inch (0.13 mm) for three place decimals and ±.01 inch (0.25 mm) for two place decimals.

FIGURE 2. Recommended mounting dimensions.

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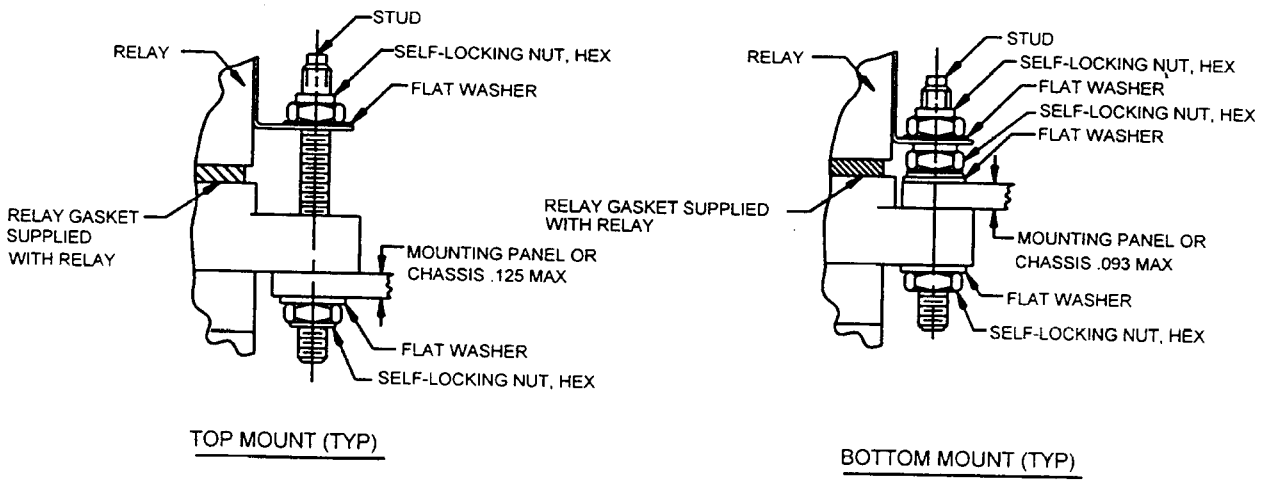


FIGURE 3. Socket mounting, fixed and loose studs.

Inches	mm
.005	0.13
.01	0.25
.093	2.36
.112	2.84
.125	3.18
.190	4.83
.22	5.6
.245	6.22
.270	6.85
.290	7.37
.318	8.08
.350	8.89

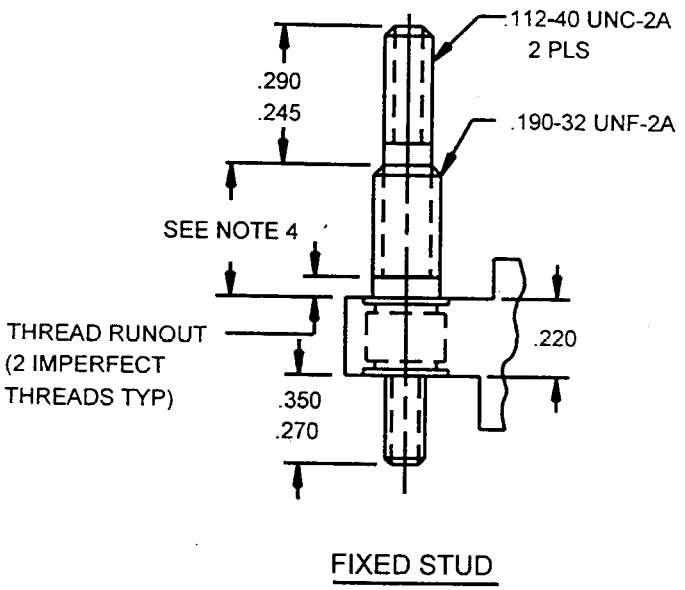
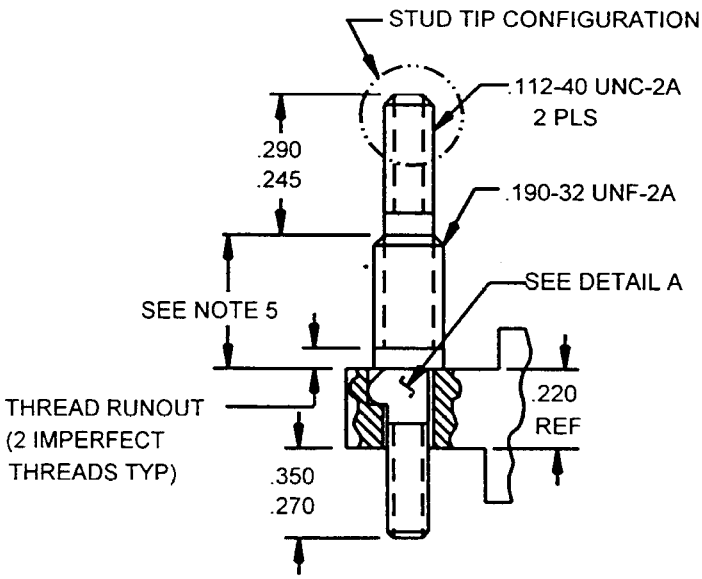
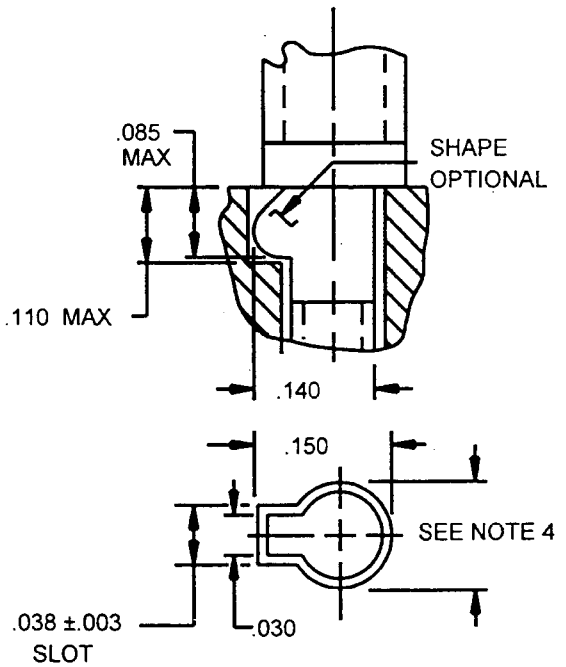


FIGURE 4. Studs fixed and loose.

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LOOSE STUD



DETAIL A

SOCKET BODY MOUNTING
HOLE AND MATING STUD

Inches	mm	Inches	mm
.003	0.08	.125	3.18
.005	0.13	.140	3.56
.01	0.25	.150	3.81
.030	0.76	.190	4.83
.038	0.97	.22	5.6
.085	2.15	.245	6.22
.093	2.36	.270	6.86
.100	2.54	.290	7.37
.112	2.84	.318	8.08
.116	2.95	.350	8.89

NOTES

1. Dimensions are in inches.
2. Metric equivalents are for general information only.
3. Unless otherwise specified, tolerances are ±.005 inch (0.13 mm) for three place decimals and ±.01 inch (0.25 mm) for two place decimals.
4. The diameter for the loose stud mating hole in the socket body shall be;
   for -01 through -06 diameter shall be .125 ±.003 inch (8.08 ±0.25 mm),
   for -11 through -15 diameter shall be .116 ±.003 inch (2.95 ±0.08 mm).
5. .290 inch (7.37 mm) for -01, -02, 04, -05 and .318 inch (8.08 mm) for -07, -08, -10, -11 sockets.

FIGURE 4. Studs fixed and loose - Continued.

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REQUIREMENTS

Design and construction: See figures 1, 2, 4, and table I.

Reference material: (see 3.1.1 of MIL-PRF-12883)

Insulator: Diallyl phthalate, in accordance with MIL-M-14, type SDG-F, any glass filled thermoplastic material in accordance with MIL-P-46179, or an approved equivalent material manufactured to a commonly accepted industry standard.

Color: Material color shall be optional providing that the color provides a contrasting background for the blue sealing grommet or blue color bands indicating rear release contacts.

Grommet: Silicon rubber in accordance with ZZ-R-765.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Electrical:

Insulation resistance: 1000 megohms minimum. Test pin diameter, size 16, .0625 ±.0010 inch (1.595 ±0.025 mm).

Dielectric withstanding voltage:

Sea level: Test voltage, 1500 V rms. Test pin diameter, size 16, .0625 ±0.0010 inch (1.595 ±0.025 mm).

High altitude (80,000 feet (24,384 meters): Test voltage, 500 V rms. Test pin diameter, size 16 .0625 ±0.0010 inch (1.595 ±0.025 mm). For purposes of this test an air pressure of 26 millibar (2.6 kilopascals), will be used to simulate an altitude of 80,000 feet (24,384 meters).

Contacts: Contacts shall be removable crimp type in accordance with MIL-C-39029/92, MIL-C-39029/5 (see table I), or approved equivalent contacts manufactured to a commonly accepted industry standard.

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TABLE I. Dash numbers and configurations.

Dash number	Mounting style	Contact size		Number of contacts	Contact designation M39029	Mating relay	Current type
		Mating end	Wire barrel				
01	Loose stud (figure 4)	16	16	20	/92 - 533	M83536/25-002, /26-002	DC
02	Loose stud (figure 4)	16	16	20	/92 - 533	M83536/27-002	AC
04	Fixed stud (figure 3)	16	16	20	/92 - 533	M83536/25-002, /26-002	DC
05	Fixed stud (figure 3)	16	16	20	/92 - 533	M83536/27-002	AC
07	Loose stud (figure 4)	16	16	20	/5 - 116	M83536/25-002, /26-002	DC
08	Loose stud (figure 4)	16	16	20	/5 - 116	M83536/27-002	AC
10	Fixed stud (figure 3)	16	16	20	/5 - 116	M83536/25-002, /26-002	DC
11	Fixed stud (figure 3)	16	16	20	/5 - 116	M83536/27-002	AC

Environmental:

Temperature range: Operating temperature range -70°C to +125°C.

Wire sealing: A resilient grommet is permanently bonded to the wire entry face of the socket so as to provide moisture sealing capabilities of from .065 inch (1.37 mm) min diameter to .1090 inch (2.29 mm) max diameter insulated wire.

Mechanical:

Vibration (sinusoidal): In accordance with MIL-STD-202, method 204, test condition G, or an equivalent industry standard.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 Hz and 3,000 Hz.

b. Except that the procedure of method 201 of MIL-STD-202 may be applied during 10 Hz to 55 Hz band of the vibration frequency range.

c. Qualified mating relay shall be used as a test gauge.

Vibration (random): In accordance with MIL-STD-1344, method 2005, test condition V, letter G, or an equivalent industry standard, with a test duration 15 minutes. The mating relay shall be used as the test gage.

Shock (mechanical): In accordance with MIL-STD-202, method 213, condition C, or an equivalent industry standard, except peak value shall be 200 g's.

Insertion and withdrawal forces: The insertion and withdrawal forces of the relay and socket shall be as specified as in table II.

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TABLE II. Insertion and withdrawal forces.

Inspection	Insertion force	Withdrawal force
Initial	30 lbf (133.45 newton) maximum	2.0 lbf (8.90 newton) minimum
After 10 insertions and withdrawals; before vibration	33 lbf (146.79 newton) maximum	1.9 lbf (8.45 newton) minimum
After vibration	33 lbf (133.45 newton) maximum	1.9 lbf (8.45 newton) minimum

Mounting hardware: The mounting hardware shall allow mounting the socket above, or below the panel or chassis (see figures 3 and 4), and shall allow mounting and securing the relay to the socket without disturbing the mounted socket or access to the wiring side of the socket. The hardware shall provide the nominal spacing between socket surface and relay mounting flange, regardless of mounting configuration.

Supplied with relay socket:

8 each .112-40 self locking nuts (.206 max dia x .176 max height).

8 each .112 flat washers (.220 max O.D. x .018 max thick).

4 each .190-32 self locking nuts (.330 max dia x .188 max height).

4 each .190 flat washers (.360 max O.D. x.019 max thick).

4 each studs for loose mounting configuration only (see figure 4).

Fixed mounting studs: Studs shall be fixed into the mounting flange of the socket and shall be designed so as to prevent rotation of the stud within the flange (see figure 3).

Contact installation tools: see table III, approved equivalent industry standard tools may also be used where appropriate.

TABLE III. Contact installation tools.

Nomenclature	Part or Identifying Number (PIN)	
Contact type	M39029/5-116	M39029/92-533
Crimp tool	M22520/1-01, /7-01	M22520/1-01, /7-01
Positioner	M22520/1 /7-03	M22520/1-02, /7-03
Insertion tool Removal tool Wired Wired	M81969/8-07 M81969/8-08, /14-03 M81969/30-06	M81969/8-07, /14-03 M81969/8-08, /14-03 M81969/30-06, /30-05

Torque: Relay socket and hardware shall be subjected to torque testing as specified in table IV. Socketsshall be installed in mounting panel when test torque is applied. No visual evidence of physical damage shall be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged (see table IV).

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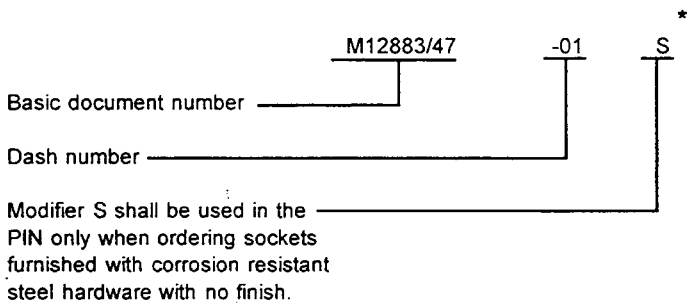
TABLE IV. Torque requirements (installed in panel conditions).

Thread size	Torque			
	Testing		Installation	
	Inch-pounds	Newton-meters	Inch-pounds	Newton-meters
.112-40	8 + 1 - 0	0.90 + .11 - 0	4 ±1	0.45 ±.11
.190-32	24 + 1 - 0	2.71 + .11 - 0	18 ±1	2.03 ±.11

Weight: .176 pound (80 grams) maximum weight of relay socket, all contacts, and all associated hardware.

PIN: The PIN shall be marked on the socket body as shown in the example (see figure 1). The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example



* For future acquisition of these sockets as of the effective date of revision D, 20 December 1989, parts identified with an "S: modifier shall be corrosion resisting steel (CRS), and parts without an "S" modifier shall be cadmium chromate finish. No mixing of hardware types shall be permitted.

Ordering data: Sockets without contacts may be ordered when so indicated in the ordering data (see MIL-PRF-12883). This applies only to original equipment manufacturers (OEM's) and subcontractors. All direct shipments to the government shall include all applicable contacts and mounting hardware. The PIN to be marked on the socket shall be as shown in the PIN example (see figure 1 and table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

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The Government PIN, specified in table V, supersedes the following commercial part numbers.

TABLE V. Supersession and cross reference.

Active Government PIN	Superseded part number
	CAGE 58982
M12883/47-01	RSE116755
M12883/47-02	RSE116757
M12883/47-04	RSE116761
M12883/47-05	RSE116763
M12883/47-07	RSL116101
M12883/47-08	RSL116103
M12883/47-10	RSL116073
M12883/47-11	RSL116075

CONCLUDING MATERIAL

Custodian:
ARMY - CR
NAVY - EC
Air Force - 85

Review activity:
Army - AR, AT, AV, ME
Navy - AS, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - ES

(Project 5935-4015-09)

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